

WEB3

Complete Learning Roadmap

Build user-facing decentralized apps that integrate wallets, contracts, and web clients.

Wallet and identity basics

Decentralized app architecture

Chain interaction model

Transaction UX

Token-based permissions

dApp security basics

01

Fundamentals

02

Core Concepts

03

Tools and Frameworks

04

Real Projects

05

Advanced Topics

5

Learning Stages

20+

Core Topics

10+

Projects

100%

Web3

Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Wallet and identity basics •
Decentralized app architecture
• Chain interaction model •
Transaction UX

02

Core Concepts

Token-based permissions •
dApp security basics •
On-chain and off-chain data
patterns • RPC fundamentals

03

Tools and Frameworks

ethers.js/wagmi

04

Real Projects

Wallet-auth app • NFT
explorer • Governance
dashboard • Decentralized
marketplace prototype

05

Advanced Topics

Gas optimization strategy •
Cross-chain dApp architecture
• Robust fallback RPC design •
User safety patterns

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Wallet and identity basics
- Decentralized app architecture
- Chain interaction model
- Transaction UX

Tools & Frameworks

- Wallet and identity basics
- Decentralized app architecture
- Chain interaction model
- Transaction UX

Recommended Projects

- Web3 workshop using Wallet and identity basics
- Web3 portfolio project with Decentralized app architecture

Stage 1 Milestone Complete Web3 workshop using Wallet and identity basics and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Token-based permissions
- dApp security basics
- On-chain and off-chain data patterns
- RPC fundamentals

Tools & Frameworks

- Token-based permissions
- dApp security basics
- On-chain and off-chain data patterns
- RPC fundamentals

Recommended Projects

- Web3 workshop using Token-based permissions
- Web3 portfolio project with dApp security basics

Stage 2 Milestone Complete Web3 workshop using Token-based permissions and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- ethers.js/wagmi
- Wallet connectors
- Indexing tools
- Contract interaction SDKs

Tools & Frameworks

- ethers.js/wagmi

Recommended Projects

- Web3 workshop using ethers.js/wagmi
- Web3 portfolio project with Wallet connectors

Stage 3 Milestone Complete Web3 workshop using ethers.js/wagmi and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Wallet-auth app
- NFT explorer
- Governance dashboard
- Decentralized marketplace prototype

Tools & Frameworks

- Wallet-auth app
- NFT explorer
- Governance dashboard
- Decentralized marketplace prototype

Recommended Projects

- Web3 workshop using Wallet-auth app
- Web3 portfolio project with NFT explorer

Stage 4 Milestone Complete Web3 workshop using Wallet-auth app and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

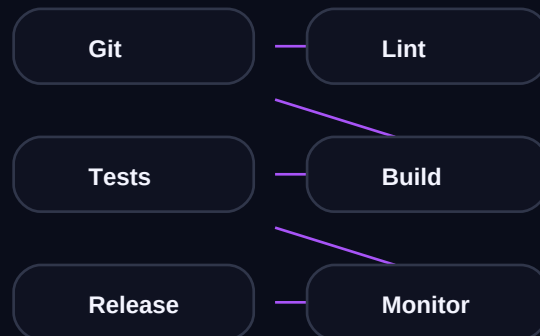
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Gas optimization strategy
- Cross-chain dApp architecture
- Robust fallback RPC design
- User safety patterns

CI/CD Pipeline



Stage 5 Milestone

A production-grade web3 workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Web3 docs and community guides

Link: <https://docs.ethers.org/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/web3>

PAID

Structured specialization track focused on Web3

Link: <https://roadmap.sh/web3>

FREE

Web3 roadmap reference

Link: <https://roadmap.sh/web3>

FREE

Official documentation

Link: <https://docs.ethers.org/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Web3?
- Which project best proves Stage 4 competency in Web3?
- What trade-offs appear in Stage 5 decisions for Web3?